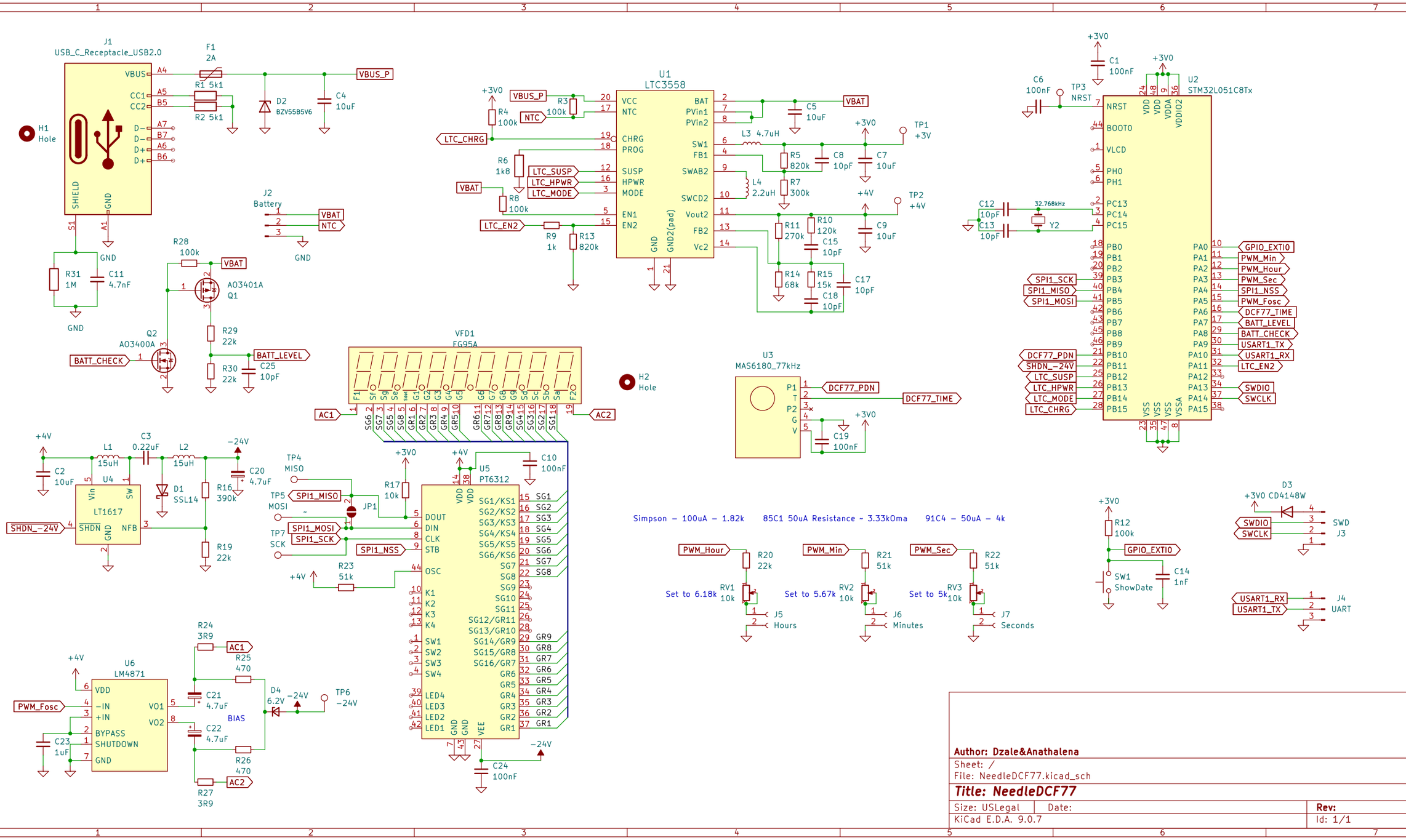

J1
USB_C_Receptacle_USB2.0
VBUS
CC1
CC2
D−
D−
D+
D+
SHIELD
GND
F1
2A
R1 5k1
R2 5k1
D2
BZV55B5V6
C4
10uF
VBUS_P
H1
Hole
J2
Battery
VBAT
NTC
GND
S1
A1
R31
1M
C11
4.7nF
R28
100k
AO3401A
Q1
Q2
AO3400A
BATT_CHECK
R29
22k
R30
22k
C25
10pF
BATT_LEVEL
U1
LTC3558
VCC
NTC
CHRG
PROG
SUSP
HPWR
MODE
EN1
EN2
GND
GND2(pad)
BAT
PVin1
PVin2
SW1
FB1
SWAB2
SWCD2
Vout2
FB2
Vc2
R3
100k
R4
100k
R6
1k8
R8
100k
R9
1k
R13
820k
L3 4.7uH
L4
2.2uH
R5
820k
R7
300k
R10
120k
R11
270k
R14
68k
R15
15k
C5
10uF
C7
10uF
C8
10pF
C9
10uF
C15
10pF
C17
10pF
C18
10pF
LTC_CHRG
LTC_SUSP
LTC_HPWR
LTC_MODE
LTC_EN2
TP1
+3V
TP2
+4V
U2
STM32L051C8Tx
NRST
BOOT0
VLCD
PH0
PH1
PC13
PC14
PC15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
VDD
VDDA
VDDIO2
VSS
VSSA
C1
100nF
C6
100nF
TP3
NRST
C12
10pF
C13
10pF
32.768kHz
Y2
SPI1_SCK
SPI1_MISO
SPI1_MOSI
GPIO_EXTI0
PWM_Min
PWM_Hour
PWM_Sec
SPI1_NSS
PWM_Fosc
DCF77_TIME
BATT_LEVEL
BATT_CHECK
USART1_TX
USART1_RX
LTC_EN2
SWDIO
SWCLK
DCF77_PDN
SHDN_−24V
LTC_SUSP
LTC_HPWR
LTC_MODE
LTC_CHRG
VFD1
FG95A
AC1
AC2
H2
Hole
U3
MAS6180_77kHz
P1
T
P2
G
V
DCF77_PDN
DCF77_TIME
C19
100nF
L1
15uH
L2
15uH
C3
0.22uF
U4
LT1617
Vin
SW
SHDN
GND
NFB
C2
10uF
C20
4.7uF
D1
SSL14
R16
390k
R19
22k
SHDN_−24V
TP4
MISO
TP5
MOSI
TP7
SCK
SPI1_MISO
SPI1_MOSI
SPI1_SCK
SPI1_NSS
JP1
R17
10k
R23
51k
U5
PT6312
DOUT
DIN
CLK
STB
OSC
K1
K2
K3
K4
SW1
SW2
SW3
SW4
LED4
LED3
LED2
LED1
SG1/KS1
SG2/KS2
SG3/KS3
SG4/KS4
SG5/KS5
SG6/KS6
SG7
SG8
SG9
SG10
SG11
SG12/GR11
SG13/GR10
SG14/GR9
SG15/GR8
SG16/GR7
GR6
GR5
GR4
GR3
GR2
GR1
C10
100nF
C24
100nF
U6
LM4871
VDD
−IN
+IN
BYPASS
SHUTDOWN
GND
VO1
VO2
BIAS
C21
4.7uF
C22
4.7uF
C23
1uF
R24
3R9
R25
470
R26
470
R27
3R9
D4
6.2V
TP6
−24V
Simpson − 100uA − 1.82k
85C1 50uA Resistance ~ 3.33kOma
91C4 − 50uA − 4k
PWM_Hour
PWM_Min
PWM_Sec
R20
22k
R21
51k
R22
51k
RV1
10k
RV2
10k
RV3
10k
Set to 6.18k
Set to 5.67k
Set to 5k
J5
Hours
J6
Minutes
J7
Seconds
R12
100k
SW1
ShowDate
C14
1nF
D3
CD4148W
SWD
J3
J4
UART
Author: Dzale&Anathalena
Sheet: /
File: NeedleDCF77.kicad_sch
Title: NeedleDCF77
Size: USLegal
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 1/1